MCQ 1: understand and choose

Five teaching questions with a single answer. Choose one answer per question; then explain each distractor. This chapter format does not reproduce the official format of the EAM.

1. $u_n = 3n + 2$: u_4 is
   - A. 14
   - B. 18
   - C. 12
   - D. 6
2. $u_0 = 2$, $u_{n+1} = u_n + 5$: u_3 is
   - A. 10
   - B. 12
   - C. 17
   - D. 15
3. An increase of 8% gives the common ratio
   - A. 0,08
   - B. 1,8
   - C. 0,92
   - D. 1,08
4. From u_3 to u_9 included, we count
   - A. 6
   - B. 7
   - C. 9
   - D. 12
5. $1 + 2 + \cdots +$ 20 is
   - A. 200
   - B. 420
   - C. 210
   - D. 190

MCQ 2: understand and choose

Five teaching questions with a single answer. Choose one answer per question; then explain each distractor. This chapter format does not reproduce the official format of the EAM.

1. $u_n = -3 \times 2^n$ is
   - A. strictly increasing
   - B. constant
   - C. neither increasing nor decreasing
   - D. strictly decreasing
2. 1 + 2 + 4 + 8 equals
   - A. 15
   - B. 16
   - C. 14

D. 8

3. For $u_n = \dfrac{16}{2^n}$, first index where $u_n < 2$:

A. 2

B. 3

C. 4

D. 5

4. The sequence $(-1)^n$

A. tends to 0

B. has no limit

C. is increasing

D. is constant

5. A list contains u_0 to u_{10} ; its length is

A. 10

B. 9

C. 12

D. 11

True or false? Justify.

1. Every geometric sequence with common ratio $q > 1$ is increasing.

2. If two sequences have equal first three terms, they are equal.

3. For $q = 1$, $1 + q + \cdots + q^n = n + 1$.

Personal review

I distinguish explicit and recursive; I recognise a constant increment or rate; I count the terms; I check the index before a threshold.
For each objective, indicate: mastered without help; achieved with help; needs more work. Write an example and a mistake to avoid.